

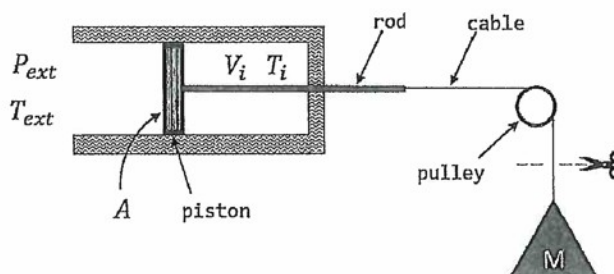
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Thermodynamics, Partial Exam, 2023-24

Do the calculations on separate sheets. Write your results in the boxes, but justify your answer by summarizing the calculations you did, so that if the result is not correct, we can evaluate your procedure and intermediate results. No results without justification will be accepted.

Problem 1 (4 points)

A monatomic gas is in equilibrium inside a container of adiabatic walls and an adiabatic piston, which separate the gas from the surroundings, kept at constant pressure P_{ext} of 1 atm and temperature T_{ext} of 25°C. The piston has an area A of 100 cm² and is connected to a rod as shown in the figure. The rod is linked to a cable which is initially connected to a mass M of 20 Kg as shown. The piston, rod, and pulley are frictionless. The gas initial volume V_i is 2 liters and its initial temperature T_i is 100°C. Gas constant $R = 8.314 \text{ J/(K}\cdot\text{mol)}$. Acceleration of gravity $g = 9.81 \text{ m/s}^2$.



1. Find the pressure P_i at this initial state. [1 point]

Since we are in equilibrium, $P_{ext} + P_{piston} = P_i$; $P_{ext} + \frac{M \cdot g}{A} = P_i$; $P_i = 101.325 \text{ Pa} + \frac{20 \text{ kg} \cdot 9.81 \text{ m/s}^2}{0.01 \text{ m}^2} = 1,209 \cdot 10^5 \text{ Pa}$

$P_i (\text{Pa}) = 1,209 \cdot 10^5$

2. If the cable is cut, the gas will evolve to a new equilibrium state. Compute the work W and the heat Q involved in the change, as well as the internal energy change ΔU , and the change in molar entropy. Assume that the gas behaves like an ideal gas. [3 points]

Since the walls are adiabatic $\Rightarrow$ adiabatic expansion against constant pressure
 $\checkmark Q = 0 \Rightarrow \Delta U = W$ $W = -P_{ext}(V_f - V_i)$ $\checkmark$

Monatomic gas $\Rightarrow \Delta U = \frac{3}{2} NR(T_f - T_i)$ Final equilibrium state $\Rightarrow P_f = P_{ext} \Rightarrow P_f V_f = NRT_f \Rightarrow V_f = \frac{NRT_f}{P_{ext}}$

$P_i V_i = NRT_i \Rightarrow N = \frac{P_i V_i}{RT_i}$. From here, we have everything expressed in terms of the initial conditions except for V_f and T_f , which we can find by solving the following system:

$\left\{ \begin{array}{l} \Delta U = \frac{3}{2} NR(T_f - T_i) = W = -P_{ext}(V_f - V_i) \\ V_f = \frac{NRT_f}{P_{ext}} \end{array} \right\}$ By developing the system we obtain an expression for T_f (also using $N = \frac{P_i V_i}{RT_i}$): $T_f = \frac{2}{5} \frac{P_{ext} T_i}{P_i} + \frac{3}{5} T_i = 348,94 \text{ K}$ $\checkmark$

$W (\text{J}) = -23,547$

$Q (\text{J}) = 0$

$\Delta U (\text{J}) = -23,547$

$\Delta S (\text{J/(K}\cdot\text{mol)}) = 7,59 \cdot 10^{-2}$

From there we can easily compute $V_f = 2,232 \cdot 10^{-3} \text{ m}^3$, and, consecutively, the Work and ΔU

For ΔS we compute an isothermal-isochoric path. Which for a monatomic ideal gas returns the following expression:

$\Delta S = R \ln \left(\frac{V_f}{V_i} \right) + \frac{3}{2} R \ln \left(\frac{T_f}{T_i} \right) = 7,736 \cdot 10^{-2} \text{ J/(K mol)}$ $\checkmark$

(To develop this expression we used that for a monatomic gas, $C_V = \frac{3}{2} R$)

Problem 2 (3 points)

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Consider 1 mol of an ideal gas undergoing the following cyclic process through states $a \rightarrow b \rightarrow c \rightarrow d \rightarrow a$:

Stage 1: a to b : Isothermal expansion at temperature T_H

Stage 2: b to c : Isochoric cooling from T_H to T_C

Stage 3: c to d : Isothermal compression at temperature T_C

Stage 4: d to a : Isochoric heating from T_C to T_H

Inputs: $T_H = 100^\circ\text{C}$; $T_C = 0^\circ\text{C}$; $c_v = 20.8 \text{ J/(K}\cdot\text{mol)}$; $V_a = 5 \text{ liters}$, $V_b = 20 \text{ liters}$. Assume reversibility.

1. Compute the heat transferred to the system in each stage of the cycle. [1 point]

In stages 1 & 3. Since the process is isothermal, and the gas is ideal we have $\Delta U(T)_1 = \Delta U(T)_3 = 0$
This implies $Q_1 = -W_1$ & $Q_3 = -W_3$

For Q_1 & Q_3 , W_1 & W_3 are obtained by the expression of work in a reversible isothermal expansion
 $Q_1 = -W_1 = -(-NRT_H \ln(\frac{V_b}{V_a})) = 4.300,80 \text{ J}$ $Q_3 = -W_3 = -(-NRT_C \ln(\frac{V_a}{V_b})) = -3.148,23 \text{ J}$ (expansion, compression)

For Q_2 & Q_4 we use the expression for heat at constant volume:

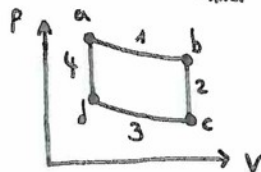
$$Q_2 = (T_C - T_H) c_v \cdot N = -2.080,00 \text{ J} \quad Q_4 = (T_H - T_C) c_v \cdot N = 2.080,00 \text{ J}$$

$$Q_1(\text{J}) = 4.300,80$$

$$Q_2(\text{J}) = -2.080,00$$

$$Q_3(\text{J}) = -3.148,23$$

$$Q_4(\text{J}) = 2.080,00$$



2. What is the net work performed on the system in one cycle, W . What is the entropy change ΔS in one cycle? [1 point]

$W_2 = W_4 = 0 \text{ J}$ since the volume is constant.

$$W = W_1 + W_3 = -Q_1 - Q_3 = -4.300,80 \text{ J} + 3.148,23 \text{ J} = -1.152,57 \text{ J}$$

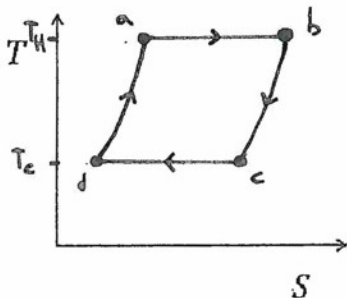
Since this process is a reversible cycle, it can be decomposed into differential Carnot cycles, therefore $\Delta S = 0 \text{ J/K}$.

$$W(\text{J}) = -1.152,57$$

$$\Delta S(\text{J/K}) = 0$$

Yes, but the reason is that S is a funct. of state. Thus, $\Delta S = 0$ in a cycle.

3. Sketch the cycle in the T - S plane below: [1 point]



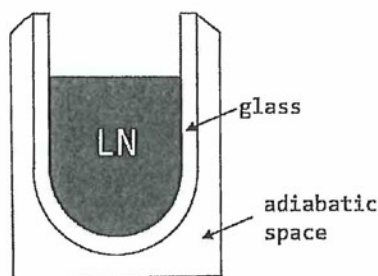
Problem 3. (3 points)

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* $T_i(\text{LN}) = 77 \text{ K}$

* $T_i(\text{glass}) = -10^\circ \text{C} = 263,15 \text{ K}$

We have a cryogenic flask for holding liquid nitrogen (LN). At the ambient pressure of the problem (1 atm) the boiling temperature of liquid nitrogen is T_b 77 K. The inner wall of the flask is made of glass, and have a mass M_w of 100 g. Initially the flask is empty and the walls' temperature is $T_i = -10^\circ \text{C}$. At one point, 2 liters of LN are poured into the flask. After reaching equilibrium, some of the LN has evaporated. Heat capacity of glass c_p is $0.780 \text{ J/(g}\cdot\text{K)}$, the latent heat of vaporization of LN h_v is 200 J/g , and the density of LN ρ_{LN} is 806 Kg/m^3 .



1. After thermal equilibrium is reached between the glass wall and the liquid nitrogen (LN), how much LN will have evaporated? [1.5 points]

We suppose less than total LN is evaporated, if the results don't add up, we will change our supposition. This implies that all of the glass reaches 77K. So let's calculate the enthalpy of this cooling process, which should be < 0 .

Since $P = \text{ct.} \rightarrow dH|_P = dQ|_P \rightarrow \Delta H_{\text{cooling}} = (T_b - T_i) M_w c_p = (77 \text{ K} - 263,15 \text{ K}) 100 \text{ g} \cdot 0,780 \text{ J/(g}\cdot\text{K)} = -14,520 \cdot 10^3 \text{ J}$

Since the recipient is an adiabatic space, we have $\Delta H = 0 \Rightarrow \Delta H_{\text{evaporation}} = -\Delta H_{\text{cooling}} > 0$

$\Delta H_{\text{evaporation}} = m_{\text{LN, ev}} \cdot h_v$; $m_{\text{LN, ev}} = \frac{\Delta H_{\text{evaporation}}}{h_v} = \frac{-\Delta H_{\text{cooling}}}{h_v} = 72,60 \text{ g}$

Now we just need to find the initial mass of LN (m_{LN}) to compute Δm_{LN} :

$m_{\text{LN}} = V_{\text{LN}} \cdot \rho_{\text{LN}} = 2 \text{ L} \cdot \frac{806 \text{ g}}{\text{L}} = 1612 \text{ g} \rightarrow \Delta m_{\text{LN}} = m_{\text{LN}} - m_{\text{LN, ev}} = 1612 \text{ g} - 72,60 \text{ g} = 1539,4 \text{ g}$

$\Delta m_{\text{LN(g)}} = 1539,4$ (72.6, BUT OK.)

2. Compute the entropy changes experienced by the LN and by the glass wall in the process. Is the process reversible or irreversible, and why? [1.5 point]

The process is irreversible, because there has been a sudden finite change in temperature. (not differential)

For the glass wall we have a temperature change at constant pressure, therefore:

$\Delta S_w = \int_{T_1}^{T_2} \frac{dQ_{\text{rev}}}{T} = \int_{T_1}^{T_2} \frac{C_p}{T} dT = C_p \ln\left(\frac{T_2}{T_1}\right) = M_w c_p \ln\left(\frac{T_b}{T_i}\right) = 100 \text{ g} \cdot 0,780 \text{ J/(g}\cdot\text{K)} \cdot \ln\left(\frac{77 \text{ K}}{263,15 \text{ K}}\right) = -95,86 \text{ J/K}$

For the LN, since the recipient is adiabatic, we can say that the heat released by the glass when cooling is absorbed by the LN. Therefore, we have the following:

$\Delta S_{\text{LN}}(\text{J/K}) = 188,57$

$\Delta S_w(\text{J/K}) = -95,86$

Reversible / Irreversible
(circle the correct one)

$Q_{\text{LN}} = -Q_{\text{glass}}|_P = -\Delta H_{\text{cooling}} = 14,520 \cdot 10^3 \text{ J}$

Now we compute ΔS_{LN} :

$\Delta S_{\text{LN}} = \frac{Q_{\text{LN}}}{T_b} = \frac{14,520 \cdot 10^3 \text{ J}}{77 \text{ K}} = 188,57 \text{ J/K}$

We know that T_{LN} remains constant, since it didn't entirely melt